

Vertex k -Separator Problem

Denis Cornaz¹, Fabio Furini¹, Mathieu Lacroix²

Enrico Malaguti³, A. Ridha Mahjoub¹, Sébastien Martin⁴

¹*LAMSADE, Université Paris-Dauphine, Place du Maréchal de Lattre de Tassigny,
75775 Paris, France*

²*LIPN, Université Paris 13, 99 Avenue Jean-Baptiste Clément, 93430 Villetaneuse,
France*

³*DEI, Università di Bologna, Viale Risorgimento 2, 40136 Bologna, Italy*

⁴*LCOMS, Université de Lorraine, Ile du Saulcy, 57045 Metz cedex, France*

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Abstract

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1 Introduction

Graph partitioning (GP) is a fundamental problem with applications in many areas. For this it has received within the past few years great attention from both theoretical and practical point of views. The graph partitioning problem for an undirected graph $G = (V, E)$ asks for a partition of V into k pairwise disjoint subsets with specific properties. For instance, the k -way partition divides the vertex set into k smaller components and the k -terminal way partition imposes that each element of the partition contains at least one terminal among a specified set of terminal vertices. Important classes of graph partition problems are the ones taking into consideration the equity and the uniformity of the subsets of vertices. For example, the problem of dividing the graph G into connected components of about the same size or the problem of finding subsets with as few connections as possible between them. The GP problem has important applications particularly in parallel computing, scheduling in multi-processor systems, data analysis, risk management of energy and social networks, clustering and detection of cliques in social, pathological and biological networks, image processing and VLSI physical design [?][?]. The GP problems related to these areas have been widely studied in the literature. But typically the algorithms developed try to find heuristic or approximated solutions. Also most exact methods are developed for the bipartization case and rely on the branch-and-bound framework. The aim of this paper is to propose efficient formulations and exact approaches for the vertex k -separator problem. This consists, given a graph $G = (V, E)$ and an integer $k > 1$, of

dividing V into $k+1$ blocks $V_0, \dots, V_k$ in such a way that $V_1, \dots, V_k$ are pairwise disjoint and V_0 is with minimum cardinality. Our objective is to devise cutting-plane algorithms based on deep algorithmic and polyhedral investigation of the combinatorial structure of the problem.

In this paper we study of the Vertex k -separator (VS) problem. The VS problem is a very important problem which can be defined as follows. Consider a simple undirected graph $G = (V, E)$, with $|V| = n$. A Vertex k -Separator (VKS) in G , for $k = 2$, is a subset of vertices $V_0 \subseteq V$ such that $V \setminus V_0$ can be partitioned in k subsets $V_1, \dots, V_k$, two-by-two disconnected, i.e., there doesn't exist an edge between two subset V_i and V_j where $i, j \in \{1, \dots, k\}$. If the cardinality of each subset V_i , $i \in \{1, \dots, k\}$, is at least greater than or equal to an integer $b(n)$, then V_0 is called Minimum Cardinality VKS (MCVKS) [?],[?][?]. And if the difference of cardinality between all pairs of subsets V_i , V_j , $i, j \in \{1, \dots, k\}$, is bounded by an integer q then V_0 is called Balanced VKS (BVKS). The MCVKS (resp. BVKS) Problem (MCVKS) (resp. BVKS) consists in finding a MCVKS (resp. BVKS) of minimum cardinality. Both problems are NP-hard in general. In this research project we will start studying the BVKSP.

Even if these problems have several applications in telecommunications or parallelism, only a few works exist concerning the exact solutions. The MCVKS, when $k = 2$ has been shown NP-hard in general but polynomial time solvable if the lower bound on the size of the blocks of the partition does not depend on n . A polyhedral study has been developed and a branch-and-cut algorithm has been tested.

The second variant that has been considered from an exact solution's point of view is the BVKSP which imposes a balanced constraint on $V_1, \dots, V_k$. Given a non negative integer q , these constraints imply that $|V_i| - |V_j| \leq q$ for all $i, j \in 1, \dots, k$. This problem has interesting applications in many fields, for example in parallel simulation of physical movements. The movement of a physical system (electric circuit, motor for example) can be modeled by a differential algebraic equation system. To resolve such a system using a parallel algorithm, it is necessary to decompose the Jacobian matrix of the system. This consists of determining a minimum group of variables (called interface variables) whose removal permits to reorganize the rest of the Jacobian into k (diagonal) blocks. In other words the remaining variables can be divided into k subgroups with no interaction between the elements of the different groups. The variables of the same block lead to a subproblem that will be then treated by a processor. Minimizing the number of interface variables allows maximum reduction of the interaction time between the different processors and, accordingly, energy and computing time savings. This problem therefore boils down to the separator problem in an appropriate graph associated to the Jacobian matrix of the system.

2 An Integer Linear Programming formulation

Consider $G = (V, E)$ a simple undirected graph and two integers k and q . For all vertex $v \in V$ and for all integer $i \in \{1, \dots, k\}$ let us associate a binary variable x_v^i which takes 1 if the vertex v is in V_i and 0 otherwise. Remark that if $\sum_{i \in \{1, \dots, k\}} x_v^i = 0$ then the vertex v is in V_0 .

The BVKSP is equivalent to the following integer linear program (P) .

$$(P) \quad \begin{aligned} & \max \sum_{i \in \{1, \dots, k\}} \sum_{v \in V} x_v^i \\ & \sum_{v \in V} (x_v^i - x_v^j) \leq q \quad i \neq j \in \{1, \dots, k\}, \end{aligned} \quad (1)$$

$$\sum_{i=1}^k x_v^i \leq 1 \quad v \in V, \quad (2)$$

$$x_u^i + x_v^j \leq 1 \quad i \neq j \in \{1, \dots, k\}, uv \in E, \quad (3)$$

$$x_v^i \in \{0, 1\} \quad i \in \{1, \dots, k\}, v \in V \quad (4)$$

$$(5)$$

The inequalities (1) ensure the maximum unbalance between the cardinality of all subset V_i , $i \in \{1, \dots, k\}$ cannot be exceed q . The inequalities (2) states that vertex v cannot be assigned to both subset V_i, V_j , where $i \neq j \in \{1, \dots, k\}$. The inequalities (3) prevent the endpoints of any edge to be assigned one to V_i , the other to V_j where V_i, V_j , $i \neq j \in \{1, \dots, k\}$.

The convex hull associated with the program (P) , defined as

$$P(G, k, q) = \text{conv}(\{x \in \{0, 1\}^{kn} \mid x \text{ satisfies (1) - (3)}\}).$$

If $\pi = (V_0, V_1, \dots, V_k)$ is a partition of G then the incidence vector associated with π , denoted by x^π , is given by

$$x_{v,i}^\pi = \begin{cases} 1 & \text{if } v \in V_i. \\ 0 & \text{otherwise,} \end{cases} \quad \forall v \in V \text{ and } \forall i \in \{1, \dots, k\}.$$

In the following we give directly different proposition. We add the proof in the last version.

Proposition 2.1. $P(G, k, q)$ is full dimensional if $q > 0$.

Remark that the inequalities (??) cannot define a facet of $P(G, k, q)$, since they are dominated by the inequalities (2).

Proposition 2.2. The inequalities (??) define a facet of $P(G, k, q)$.

Proposition 2.3. The inequality (1), associated with $v \in V$, defines a facet of $P(G, k, q)$ if and only if either $q > 1$ or for all edges $uv \in E$ there exists in $V \setminus \tilde{N}(\{u, v\})$ an independent set of cardinality greater or equal to $k - 1$.

The edge inequalities (??) cannot define a facet. In the following proposition we strengthen the edge inequalities.

Proposition 2.4. Let $uv \in E$ be an edge and $z \leq k$ be an integer. The inequality

$$x_u^z + \sum_{t \in \{1, \dots, k\} \setminus \{z\}} x_v^t \leq 1, \quad (6)$$

is valid.

Proposition 2.5. The inequality (6) associated with $uv \in E$ and $z \in \{1, \dots, k\}$ defines a facet of $P(G, k, q)$ if and only if either $q > 1$ or G contains an independent set of cardinality greater or equal to $k - 1$ in $V \setminus \tilde{N}(\{u, v\})$.

Now we propose an inequality family that dominates the balanced inequalities (??) given in the basic formulation.

For all $v \in V$, we denote by $\bar{N}'(v)$ the set of vertices in $V \setminus \{v\}$ such that they are not adjacent to v . $\bar{N}'(v) = V \setminus (N(v) \cup \{v\})$.

Proposition 2.6. Let $W \subseteq V$ the set of vertices such that for all $v \in W$, either $|\bar{N}'(v)| < q + k - 1$ or $\bar{N}'(v)$ doesn't contain an independent set of cardinality $k - 2$. The following inequalities

$$\sum_{v \in V} x_v^t - \sum_{u \in V \setminus W} x_u^r \leq q \quad \text{for all } t \neq r \in \{1, \dots, k\}, \quad (7)$$

are valid.

We denote these inequalities by *w-balanced inequalities*.

Now we give the necessary and sufficient conditions for these inequalities to be facet defining.

Proposition 2.7. Inequalities (7) define a facet of $P(G, k, q)$ if and only if for all vertices $v \in W$, $|\delta(v)| \leq n - q - 1$ ($\Leftrightarrow \bar{N}'(v) \geq q$).

3 Extended formulation

Let $S^t = \{s \subseteq V\}$ associated with partition $t \in \{1, \dots, k\}$. Using the following binary variable:

$$x_s \quad (s \in S^t) = \begin{cases} 1 & i \in s \text{ are in the } t\text{-th partition} \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

$$\max \sum_{t \in \{1, \dots, k\}} \sum_{s \in S^t} W_s x_s \quad (9)$$

$$\sum_{t \in \{1, \dots, k\}} \sum_{s \in S^t: v \in s} x_s \leq 1 \quad v \in V \quad (\lambda_v) \quad (10)$$

$$\sum_{t \in \{1, \dots, k\}} \sum_{s \in S^t} a_s^{uv} x_s \leq 1 \quad (u, v) \in E \quad (\pi_{uv}) \quad (11)$$

$$\sum_{s \in S^t} x_s = 1 \quad t \in \{1, \dots, k\} \quad (\gamma_k) \quad (12)$$

$$\sum_{s \in S^t} W_s x_s - \sum_{s \in S^l} W_s x_s \leq q \quad t, l \in \{1, \dots, k\}, t \neq l \quad (\rho_{tl}) \quad (13)$$

$$x_s \in \{0, 1\} \quad t \in \{1, \dots, k\}, s \in S^t \quad (14)$$

where

$$W_s = |s| \quad (15)$$

and

$$a_s^{uv} = \begin{cases} 1 & u \in s \vee v \in s \\ 0 & \text{otherwise} \end{cases} \quad (16)$$

once relaxed (14) and given $(\lambda_u^*, \gamma_t^*, \pi_{uv}^*, \rho_{tl}^*, \rho_{tl}'^*)$ the separation of the dual constraints is to find a $t^* \in \{1, \dots, k\}$, $s^* \in S^{t^*}$ such that:

$$\sum_{i \in s^*} \lambda_i^* + \sum_{(ij) \in E} a_{s^*}^{ij} \pi_{ij}^* + \gamma_{t^*}^* + W_{s^*} \sum_{l \in \{1, \dots, k\}, l \neq t^*} (\rho_{t^*l}^* - \rho_{t^*l}'^*) < W_{s^*} \quad (17)$$

Defining

$$\Omega^{t^*} = \sum_{l \in \{1, \dots, k\}, l \neq t^*} (\rho_{t^*l}^* - \rho_{t^*l}'^*)$$

The separation (17) is equivalent to:

$$\sum_{v \in s^*} (\lambda_v^* + \Omega^{t^*} - 1) + \sum_{(uv) \in E} a_{s^*}^{uv} \pi_{uv}^* + \gamma_{t^*}^* < 0 \quad (18)$$

To check this condition we can solve for each $t \in \{1, \dots, k\}$ the following MIP using a binary and a continuous variables:

$$z_v \quad (v \in V) = \begin{cases} 1 & v \in s^* \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

$$\min \sum_{v \in V} (\lambda_v^* + \Omega^{t^*} - 1) z_v + \sum_{(u,v) \in E} \pi_{uv}^* y_{uv} \quad (20)$$

$$y_{uv} \geq z_u \quad (u, v) \in E \quad (21)$$

$$y_{uv} \geq z_v \quad (u, v) \in E \quad (22)$$

$$z_v \in \{0, 1\} \quad v \in V \quad (23)$$

The (column generation) subproblem is the following: we have a graph $G = (V, E)$, a profit λ_v for each $v \in V$ and a penalty $\pi_{uv} \geq 0$ for each $(u, v) \in E$ (this summarizes the coefficients discussed in equation (20)). We want to select the subset of vertices of maximum profit; if one vertex v is selected, the penalty π_{uv} associated with any edge $(u, v) \in E$ is paid.

Without loss of generality, we may assume $\lambda_v > 0$ for each $v \in V$ (vertices with $\lambda_v \leq 0$ can be removed, their incident edge penalties go to the other endpoint) and a penalty $\pi_{uv} > 0$ for each $(u, v) \in E$ (if $\pi_{uv} = 0$ we can remove the edge).

For $V' \subseteq V$, let:

$$\begin{aligned} \lambda(V') &= \sum_{v \in V'} \lambda_v, \\ \pi(E(V')) &= \sum_{u, v \in V': (u, v) \in E} \pi_{(u, v)}, \\ \pi(\delta(V')) &= \sum_{v \in V', u \notin V': (u, v) \in E} \pi_{(u, v)}. \end{aligned}$$

The subproblem is polynomially solvable because of:

- Submodularity
- Constraint matrix is TUM
- As a max-flow problem
- The dual problem can be solved as a max-flow problem

4 Conclusion

The conclusion goes here.

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References

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